State-Transfer Protocols for Multilingual Multi-Agent Handoff

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Abstract

Large language model agents increasingly need to continue work after context truncation, provider failure, model substitution, or handoff to another agent. This paper studies compact prompt formats as *state-transfer protocols*: reusable communication formats intended to preserve operational state rather than merely reduce token count. We evaluate four modes—`natural`, `compressed`, `hybrid_min`, and `hybrid_state`—across multilingual, multi-model, judge-based, and real-agent settings. EXP05 provides a broad discovery study over Spanish, English, and Chinese tasks with multiple generator families. EXP06 is a controlled follow-up comparing `compressed` against `hybrid_state`, including language/variant ablations and three-judge calibration. EXP07 and EXP08 move beyond LLM-as-judge evaluation by testing real-agent handoff in LangGraph/OpenFang-style surfaces with objective continuation metrics. Across these experiments, `compressed` is the strongest general baseline, while `hybrid_state` improves judge-scored state preservation in paired comparisons but does not win global quality or real-agent operational performance. We argue that prompt compression should be reframed as protocol design: compact formats should be selected by target function, such as semantic quality, state preservation, continuity, cost, or interface reliability.

1 Introduction

Agentic LLM systems are often interrupted. Context windows end, workers change, provider quotas fail, and downstream agents need a compact representation of what happened. A common response is to summarize more context in natural language. That is useful, but it is not the same as preserving operational state. Variables, constraints, unfinished subtasks, plans, and forbidden claims can disappear even when the summary sounds fluent.

This paper asks whether compact prompt formats can be treated as *state-transfer protocols*. The question is not only whether a prompt can be shorter, but whether it can carry the right state for another model, language, or agent surface to continue the task. This distinction matters because the best format for global quality may differ from the best format for handoff continuity.

The main claim is deliberately narrow. `compressed` is a robust universal baseline. `hybrid_state` is not a global winner, but it is a specialized protocol that improves state preservation in judge-scored paired designs. In real-agent settings, the advantage becomes conditional: objective validators show that `compressed` often remains stronger, while `hybrid_state` can help selected constraint/plan strata.

2 Related Work

This work sits between prompt compression, agent memory, tool-use evaluation, and LLM-as-judge methodology. Prompting work such as chain-of-thought shows that format changes can alter model behavior, but the goal here is not to elicit longer reasoning; it is to transfer compact operational state across handoffs. Prompt-compression work such as LLMLingua shows that inputs can often be shortened without proportional quality loss, but our focus is not only token reduction: it is whether compact formats preserve variables, constraints, and next actions. ReAct-style systems connect language models to action traces and tools, making state representation a practical interface problem rather than only a prompt-design problem. Agent-memory systems such as Generative Agents highlight the value of persistent state, while long-context studies show that simply storing more text does not guarantee retrieval or use of the right information. Finally, LLM evaluation surveys motivate caution around judge-scored results; for this reason, EXP07 and EXP08 add objective continuation metrics and treat EXP09/EXP10 as interface validations rather than broad agent benchmarks.

3 Protocols

We evaluate four communication modes. Table 1 shows the same handoff task rendered in each style.

Mode	Example handoff for: "continue EXP06 analysis without claiming global win"
`natural`	Continue the EXP06 analysis. The current result is that `compressed` remains stronger on global quality, while `hybrid_state` improves state preservation. Do not claim that `hybrid_state` wins overall. Next, report paired deltas and limitations.
`compressed`	goal=EXP06 report; state=compressed wins Q, hybrid_state wins state; forbid="hybrid_state global winner"; evidence=paired deltas; next=write result + limitation; risk=overclaim.
`hybrid_min`	g=EXP06; st=comp>Q, hs>state; !claim=hs global win; e=paired deltas; n=report+limit; r=overclaim.
`hybrid_state`	`HYBRID_STATE{G:EXP06; VAR:[base=compressed, cand=hybrid_state]; C:[no global-win claim]; E:[Q lower, state higher]; PLAN:[paired deltas, language ablation, limits]; NEXT:bounded result}`

Table 1: Concrete protocol examples. The distinction is not only brevity: `hybrid_state` makes operational variables and constraints explicit.

4 Experimental Design

EXP05 is a broad discovery experiment. It evaluates 15 task templates across Spanish (ES), English (EN), and Chinese (ZH), four protocol modes, two repetitions, and five generator families: Grok 4.20 Non-Reasoning, Gemini 3.1 Pro Preview, Llama 4 Maverick, Qwen3-Next-80B, and GLM 5. Judges return strict JSON metrics including semantic fidelity, utility, state preservation, operational continuity, handoff quality, compactness, ambiguity, and information loss.

EXP06 is a controlled follow-up. It compares `compressed` and `hybrid_state` under paired cells, adds language/variant ablations (ES, EN, ZH, ZH_PINYIN, ES_LITERAL, EN_LITERAL), and evaluates valid generations with three judge routes: Gemini 3.5 Flash, Azure GPT 5.4 High, and Grok 4.20 Reasoning.

EXP07 and **EXP08** test real-agent handoff. They use LangGraph/OpenFang-style execution with objective metrics: variable recovery, subtask completion, plan continuity, constraint retention, handoff success, and state-error count. EXP07 contains 270 cleaned executions. EXP08 scales to 900 operational executions and reports both an operational-complete view and a strict JSON-valid view.

Exp.	Role	Key result	Interpretation
EXP05	Broad multilingual discovery	`compressed` strongest overall; `hybrid_state` improves state preservation by +0.133 vs. `compressed`	State preservation can diverge from global quality.
EXP06	Controlled paired follow-up	`hybrid_state` loses quality (−0.208) but gains state preservation (+0.066)	Confirms specialization, rejects global-win claim.
EXP07	Real-agent objective handoff	`compressed` leads variable recovery and plan continuity; `hybrid_state` only slightly higher on constraints	Judge-state gains do not automatically transfer to operational metrics.
EXP08	Real-agent scale-up	900 operational executions; strict JSON-valid view exposes 76 blank-output failures in GPT 5.4 High	Protocol behavior depends on parsers, schemas, and serving route behavior.

Table 2: Compact summary of the main evidence used in this short paper. EXP09/EXP10 are excluded from central claims because they are saturated tool-interface validations.

EXP09/EXP10 are not treated as central evidence for protocol superiority in this paper. They are interface-validation studies for tool-use and public-repository maintenance. Their saturated success rates are better read as engineering lessons about action schemas, parsers, validators, checksums, and reasoning-budget behavior.

5 Results

5.1 EXP05: Broad Discovery

Figure 1 summarizes EXP05. `compressed` is the strongest global baseline. `hybrid_state` is close on global quality and improves state preservation in paired analysis, while `hybrid_min` preserves compactness but loses quality.

The paired comparison against `compressed` shows the tradeoff directly. `hybrid_state` improves state preservation by $+0.133$ with 95% CI $[+0.025, +0.240]$, while losing global quality by -0.108 with 95% CI $[-0.203, -0.012]$. Operational continuity is effectively tied.

5.2 EXP06: Controlled Follow-Up

EXP06 confirms the state-preservation interpretation but weakens any global-win interpretation. Figure 2 shows that `compressed` remains stronger on quality, fidelity, compactness, and lower ambiguity/information loss, while `hybrid_state` is higher on state preservation.

Overall paired quality favors `compressed`. The observed `hybrid_state` minus `compressed` quality delta is -0.208, with 95% CI [-0.239, -0.177]. State preservation moves in the opposite direction: +0.066, with 95% CI [+0.032, +0.100]. This is the central controlled result.

Figure 3 shows quality deltas by language/variant. Native ZH remains more favorable than ZH_PINYIN, suggesting that representation matters, but this should be treated as a promising signal rather than a causal tokenization proof.

5.3 EXP07/EXP08: Real-Agent Objective Metrics

EXP07 and EXP08 reduce dependence on LLM judges by scoring continuation behavior directly. Figure 4 shows the compact objective-metric view.

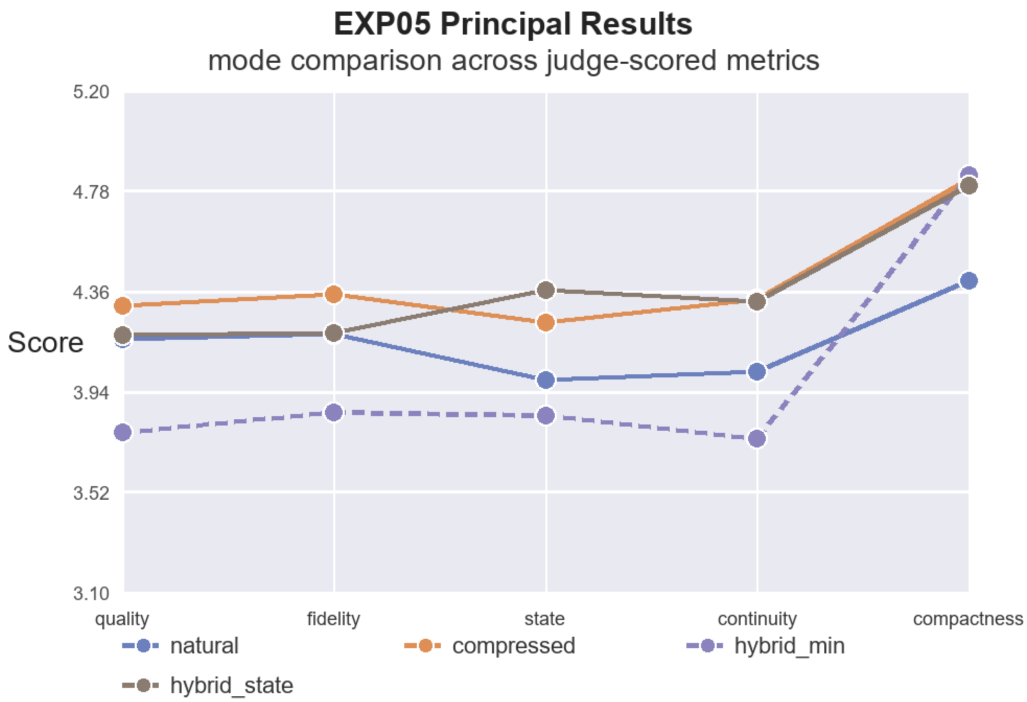

Figure 1: EXP05 principal results by protocol mode. The curve shape is the key result: compactness alone is not enough, and state preservation separates `hybrid_state` from `hybrid_min`.

`compressed` leads variable recovery and plan continuity. `hybrid_state` is only slightly higher on constraint retention, and does not dominate globally.

EXP08 scales the setting to 900 operational executions. The operational-complete view again favors `compressed` globally. The strict JSON-valid view reveals a separate model/serving failure mode: Azure GPT 5.4 High produced 76 blank-output cells under strict JSON scoring, concentrated more heavily in `hybrid_state`. This supports a systems interpretation: protocol performance depends on parser contracts, model route behavior, and output-budget settings.

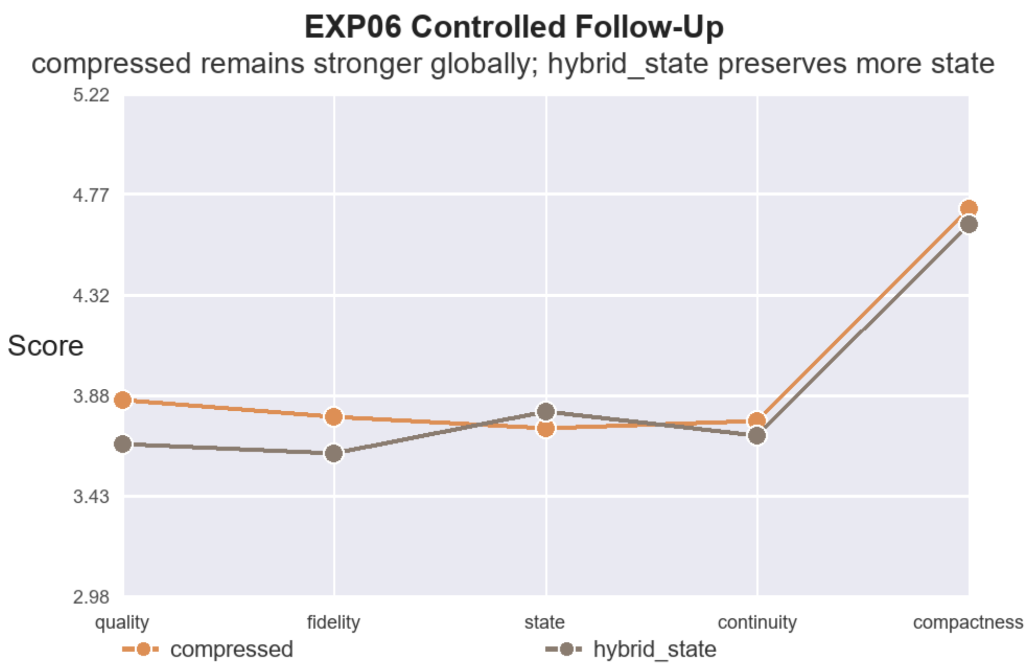

Figure 2: EXP06 controlled follow-up. `hybrid_state` is a state-oriented specialization, not a universal quality improvement.

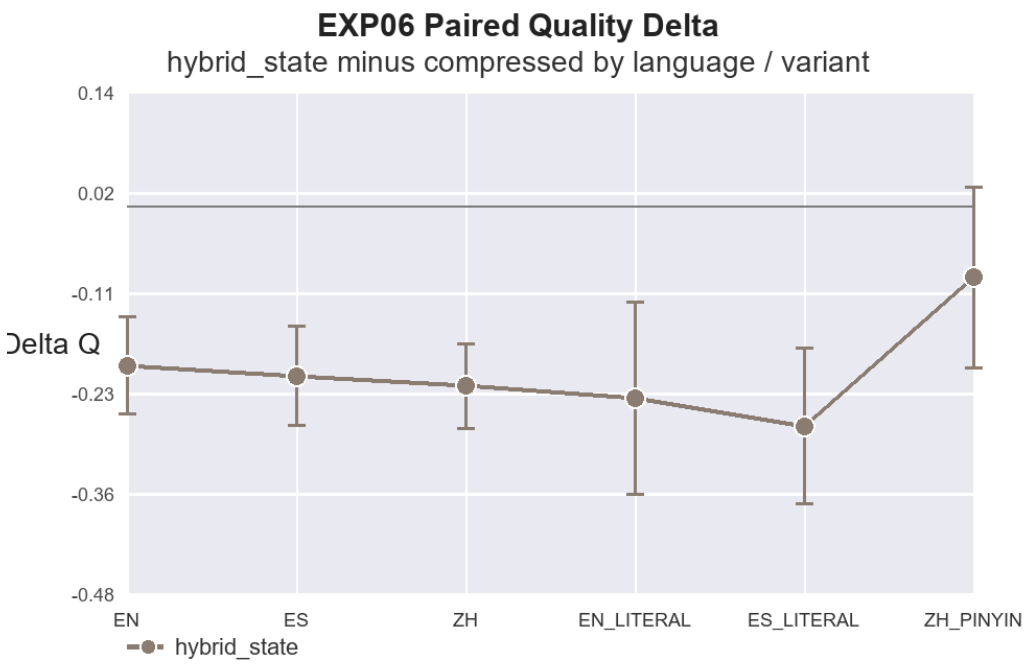

Figure 3: EXP06 paired quality delta for `hybrid_state` minus `compressed`. Negative values mean `compressed` remains stronger on global quality.

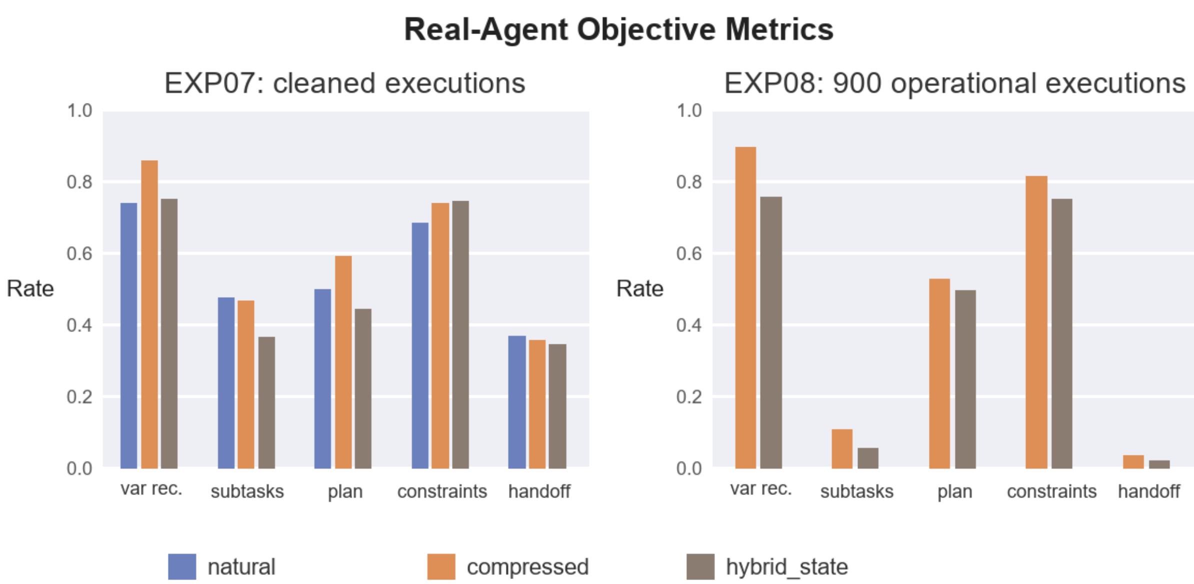

Figure 4: EXP07/EXP08 real-agent objective metrics with rate axes fixed to $[0, 1]$. The compact view reduces reliance on LLM judges while showing that `compressed` remains the stronger operational baseline in these settings.

6 Discussion

The results support a target-function view of compact protocols. If the goal is broad semantic quality and robustness, `compressed` is the safest baseline. If the goal is explicit state preservation under judge-scored handoff, `hybrid_state` is a useful specialization. If the goal is real-agent execution, the protocol alone is insufficient: schemas, parsers, validators, and model serving behavior become part of the experimental surface.

The ZH signal is interesting but not settled. EXP05 shows strong ZH performance, and EXP06 shows that native ZH is more favorable than ZH_PINYIN. This supports a representation hypothesis, but does not isolate tokenizer mechanics from training distribution, script familiarity, or provider routing.

EXP09 and EXP10 are best treated as interface lessons. Their saturated final success rates do not discriminate protocols strongly, but their repaired failures are valuable: schema mismatch, parser contract failures, checksum hallucination, missing tool names, and thinking-budget/output behavior. These are not distractions; they are exactly the kinds of failures that determine whether a state-transfer protocol survives contact with real tools.

7 Limitations

The experiments are systems-oriented and provider-dependent. Model names, routes, quotas, and serving behavior are time-sensitive. EXP05 is broad and exploratory. EXP06 is better controlled but still relies on LLM judges. EXP07 and EXP08 add objective metrics, but their framework surfaces are not interchangeable. The quality index is descriptive rather than psychometrically validated. Strong claims should therefore rely on individual metrics and paired contrasts, especially state preservation and operational continuity.

The statistical analysis uses paired deltas and confidence intervals. A stronger venue submission should add clustered bootstrap or mixed-effect models over task, language, generator, judge, and repetition. This is especially important because the design contains multiple comparisons and correlated cells.

8 Conclusion

Prompt compression should be reframed as state-transfer protocol design. Across EXP05–EXP08, `compressed` is the robust universal baseline. `hybrid_state` is a narrower state-oriented protocol: it improves judge-scored state preservation but does not win global quality or real-agent operational metrics. The most useful outcome is therefore not a single winning prompt style, but a clearer experimental vocabulary for evaluating agent handoff: semantic quality, state preservation, continuity, compactness, objective recovery, and interface reliability.

Data and Code Availability

All central claims in this short paper are supported by the long technical report and reproducibility package. The technical report contains the extended methodology, experiment history, cost notes, error analysis, and appendices. The reproducibility repository contains prompts, task banks, validators, schemas, cleaned latest-cell datasets, repaired-error logs, analysis scripts, freeze manifests, and checksum manifests. Public release should exclude API keys, credential files, private provider logs, and unredacted local paths.

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